



Object Storage with AWS S3 and GCP GCS

A comprehensive guide to leveraging cloud object storage for AI and ML workflows

Why Object Storage for AI?

Massive Scale

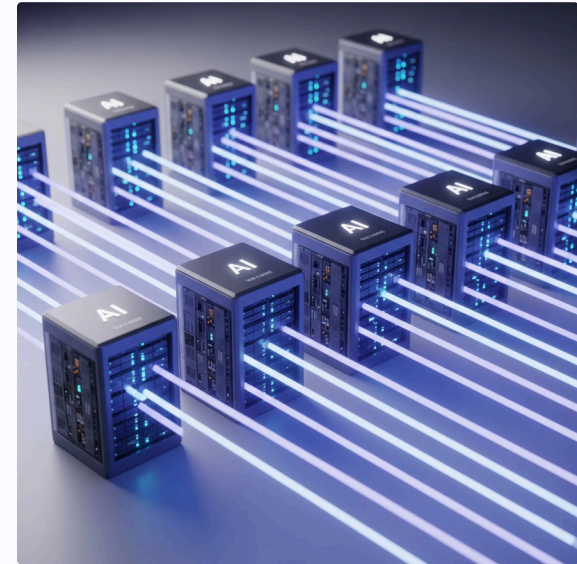
AI workloads require virtually unlimited storage capacity that can grow with your data needs

Cost-Effective

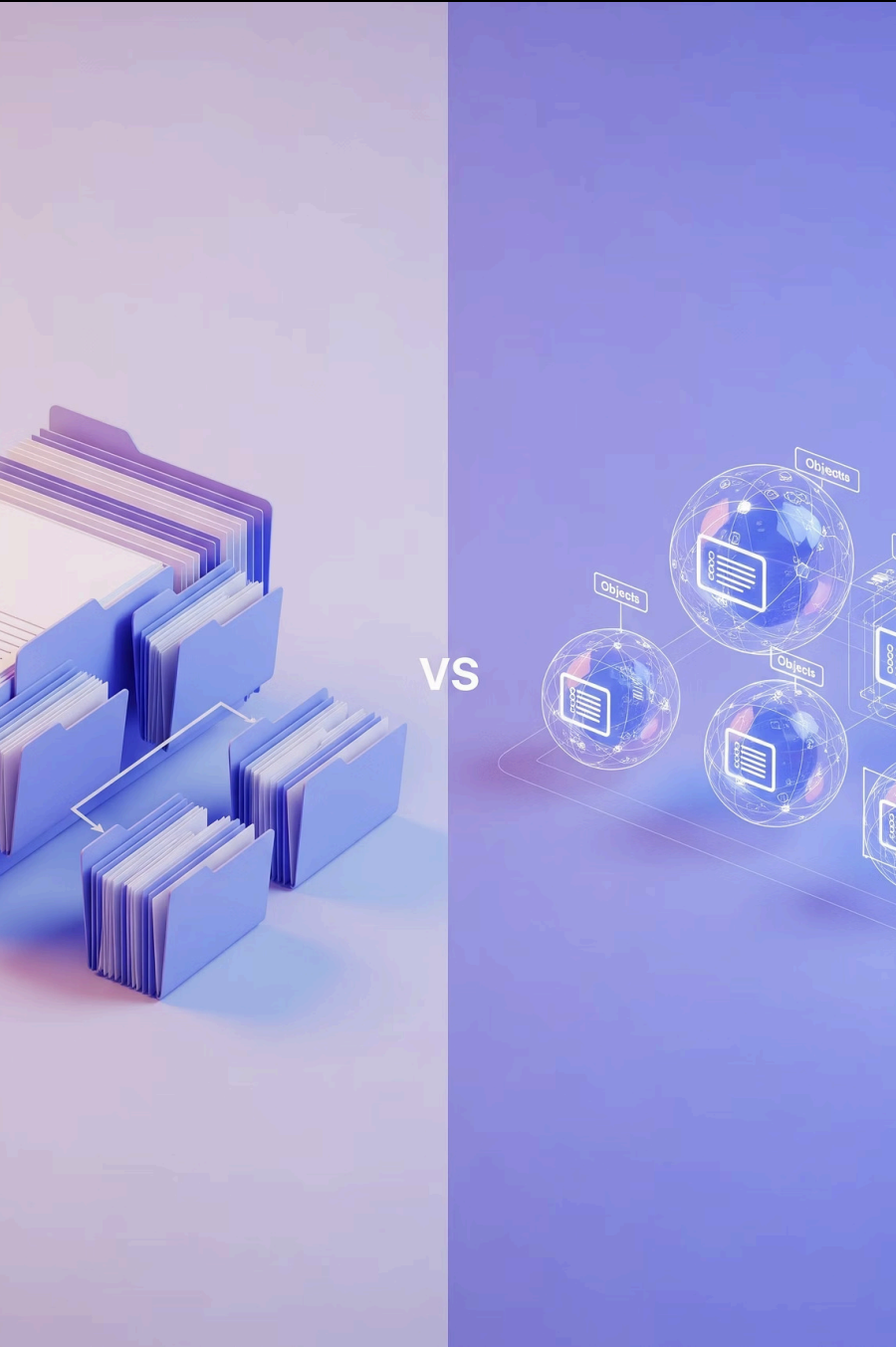
Cloud-native solution with significantly lower costs than block storage alternatives

Parallel Access

Supports high-throughput concurrent reads for distributed training across multiple nodes



Perfect for unstructured data: images, text, video, and audio files used in AI training



What Is Object Storage?

Object Structure

Each object contains data + metadata + unique identifier

No traditional file system hierarchy

Access Methods

RESTful APIs instead of file paths

HTTP/HTTPS operations (GET, PUT, DELETE)

Key Providers

AWS S3

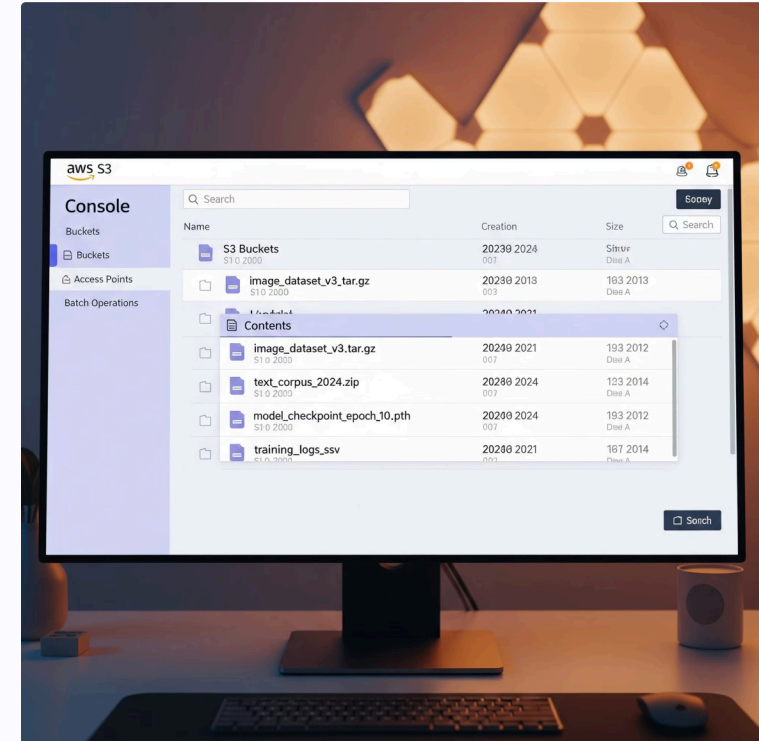
Google Cloud Storage

Azure Blob Storage

AWS S3 Overview

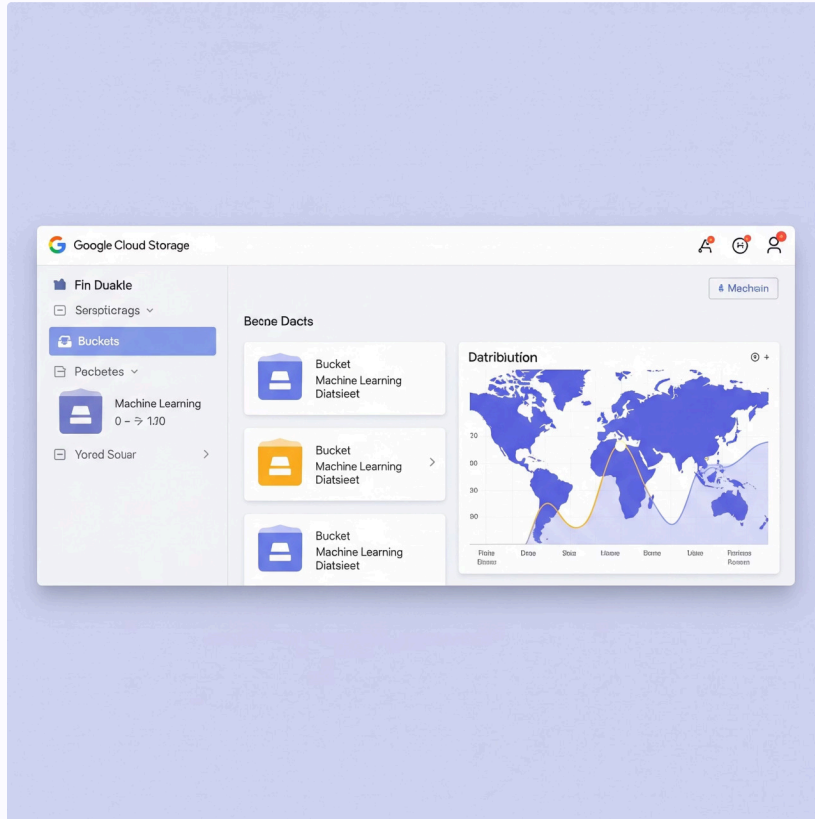
Simple Storage Service (S3) is the industry standard for cloud object storage, offering:

- Data organized in **buckets** with globally unique names
- Eleven 9's of durability (99.999999999%)
- Rich feature set: **versioning, replication, lifecycle rules**
- Native integration with AWS AI/ML services
- Multiple access methods: CLI, SDK, REST API



The backbone of AWS-based ML pipelines

GCP GCS Overview



Google Cloud Storage Features

- Global namespace with regional/multi-regional options
- Strong consistency model (immediate read-after-write)
- **Deep integration with BigQuery** for AI analytics
- Optimized for Google Cloud AI Platform
- Access via gsutil CLI, client libraries, or REST API

Perfect companion for TensorFlow and JAX workloads

Storage Classes: Balancing Cost vs Performance

1

Standard / Hot Storage

Optimized for **frequent access** with millisecond retrieval

Ideal for [active AI training datasets](#) and models in production

Highest cost per GB, no retrieval fees

2

Infrequent Access / Nearline

Lower storage costs with **moderate retrieval fees**

Good for monthly batch processing or backups of important models

Retrieval within seconds

3

Archive / Coldline

Lowest storage costs with **highest retrieval fees**

Best for regulatory compliance or long-term model preservation

Retrieval can take minutes to hours

❏ Most AI workloads use Standard/Hot storage for active training and Infrequent Access for completed projects

Security and Access Control

Identity & Access Management (IAM)

Control access by users, service accounts, and roles

Principle of least privilege: grant minimum necessary permissions

Bucket Policies & ACLs

Fine-grained rules for resource-level permissions

Control access at bucket or object level

Encryption

Server-side encryption with KMS keys

TLS encryption for data in transit



Essential for GDPR, HIPAA, and other compliance requirements

Using AWS S3 – CLI Basics

```
# Upload data to S3
aws s3 cp data.csv s3://my-bucket/data.csv

# Download from S3
aws s3 cp s3://my-bucket/data.csv .

# List bucket contents
aws s3 ls s3://my-bucket/
```

The AWS CLI makes it easy to:

- Automate data transfer in ML pipelines
- Script batch operations for dataset preparation
- Integrate with CI/CD workflows for model deployment



AWS CLI: The data engineer's best friend

Using GCP GCS – CLI Basics

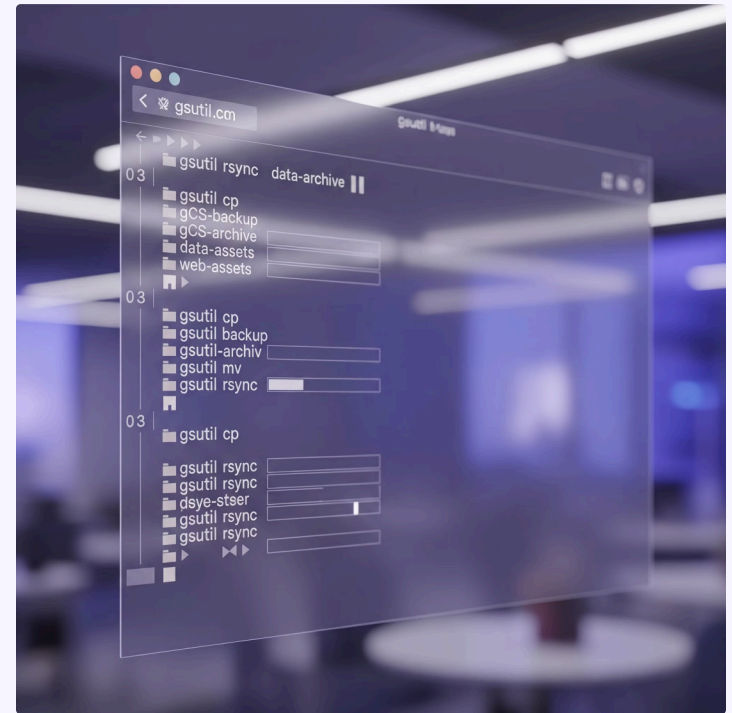
```
# Upload data to GCS
gsutil cp data.csv gs://my-bucket/data.csv

# Download from GCS
gsutil cp gs://my-bucket/data.csv .

# List contents
gsutil ls gs://my-bucket/
```

The gsutil tool provides:

- Python SDK integration for custom workflows
- Easy connection to Vertex AI and BigQuery
- Parallel composite uploads for large datasets



gsutil makes GCS operations streamlined for AI pipelines

ML Pipeline



AI Use Cases for Object Storage



Raw Datasets

Store unstructured data: images, text, video, audio



Model Checkpoints

Save training progress and enable resumability



Inference Artifacts

Host tokenizers, configs, and quantized models



Distributed Training

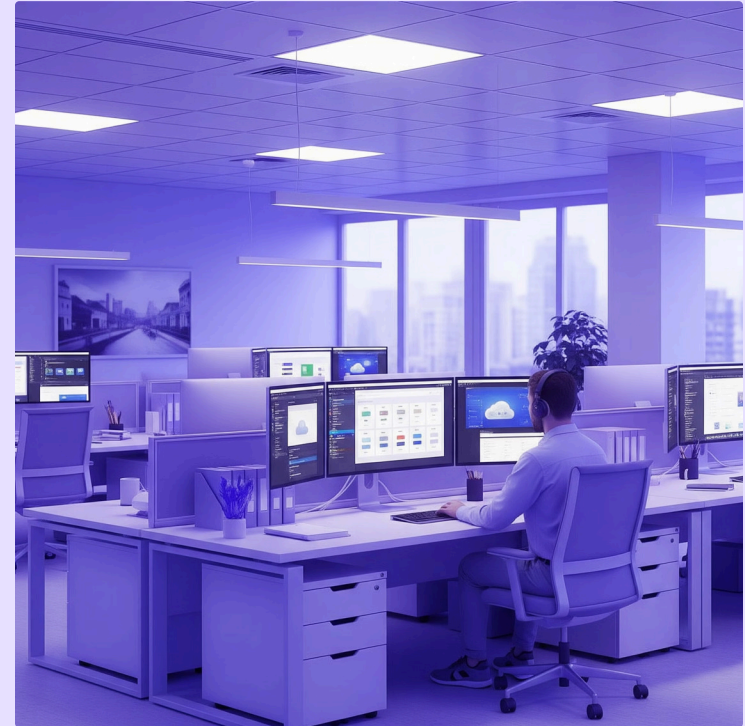
Enable cross-region replication for global workloads

Key Takeaways

Master Object Storage

Object storage is the **backbone of cloud AI pipelines**, providing the scalability and durability needed for modern machine learning workflows.

- AWS S3 and GCP GCS are industry-leading solutions
- Use storage classes strategically to optimize costs
- Implement proper security for sensitive data
- Leverage APIs and CLIs for automation



Understanding object storage is a **fundamental skill** for AI infrastructure engineers and ML practitioners